

Carbonated Soft Drinks: Formulation and Manufacture

Edited by

David P. Steen
and
Philip R. Ashurst



Blackwell
Publishing

Carbonated Soft Drinks: Formulation and Manufacture

Carbonated Soft Drinks: Formulation and Manufacture

Edited by

David P. Steen
and
Philip R. Ashurst



Blackwell
Publishing

© 2006 by Blackwell Publishing Ltd

Editorial Offices:

Blackwell Publishing Ltd, 9600 Garsington Road, Oxford OX4 2DQ, UK

Tel: +44 (0)1865 776868

Blackwell Publishing Professional, 2121 State Avenue, Ames, Iowa 50014-8300, USA

Tel: +1 515 292 0140

Blackwell Publishing Asia Pty Ltd, 550 Swanston Street, Carlton, Victoria 3053, Australia

Tel: +61 (0)3 8359 1011

The right of the Author to be identified as the Author of this Work has been asserted in accordance with the Copyright, Designs and Patents Act 1988.

All rights reserved. No part of this publication may be reproduced, stored in a retrieval system, or transmitted, in any form or by any means, electronic, mechanical, photocopying, recording or otherwise, except as permitted by the UK Copyright, Designs and Patents Act 1988, without the prior permission of the publisher.

First published 2006 by Blackwell Publishing Ltd

ISBN-13: 978-14051-3435-4

ISBN-10: 1-4051-3435-6

Library of Congress Cataloging-in-Publication Data

Carbonated soft drinks: formulation and manufacture/edited by David P. Steen and Philip R. Ashurst.
p. cm.

Includes bibliographical references and index.

ISBN-13: 978-1-4051-3435-4 (alk.paper)

ISBN-10: 1-4051-3435-6 (alk.paper)

1. Soft drinks. 2. Carbonated beverages. I. Steen, David P. II. Ashurst, P. R.

TP630.C38 2006

663'.62-dc22

2005035492

A catalogue record for this title is available from the British Library

Set in 10/12 pt Times

by Newgen Imaging Systems (P) Ltd., Chennai, India

Printed and bound in India

by Replika Press Pvt, Ltd, Kundli

The publisher's policy is to use permanent paper from mills that operate a sustainable forestry policy, and which has been manufactured from pulp processed using acid-free and elementary chlorine-free practices. Furthermore, the publisher ensures that the text paper and cover board used have met acceptable environmental accreditation standards.

For further information on Blackwell Publishing, visit our website:

www.blackwellfood.com

Contents

List of Contributors	xv
Preface	xvii
1 Introduction	1
Bob Hargitt	
1.1 Early history	1
1.2 The growth of carbonates – production	3
1.3 Technological development	4
1.3.1 Carbon dioxide	4
1.3.2 Sweeteners	5
1.3.3 Flavours and colours	7
1.3.4 Packaging	8
1.4 Recent technological development	10
1.5 The growth of carbonates – consumption	10
1.6 Changing public perception and future challenges	13
Acknowledgements	14
Bibliography	14
2 Water treatment	16
Walter Tatlock	
2.1 Introduction	16
2.2 Water quality	18
2.3 Town mains water	21
2.4 Boreholes	23
2.5 How to achieve the desired water quality	24
2.6 Sand filtration	25
2.7 Coagulation	27
2.8 Alkalinity reduction	28
2.9 Membrane filtration	29
2.10 Chlorination and carbon filtration	34
2.11 Iron removal	35
2.12 Nitrate removal	36
2.13 Polishing filters	38
2.14 Ultraviolet systems	41
2.15 Ozone technology	42

2.16	De-aeration	43
2.17	Factory water distribution systems	44
2.18	Factory water systems	46
	References	46

3 Ingredients and formulation of carbonated soft drinks 48

Barry Taylor

3.1	Introduction	48
3.2	Factors influencing development of the industry	48
3.3	The move towards standardisation	51
3.4	The constituents of a soft drink	52
3.5	Water	54
3.5.1	Requirements	54
3.5.2	Quality of fresh water	55
3.5.3	Water hardness	55
3.5.4	Water treatment	55
3.5.5	Water impurities and their effect	55
3.5.5.1	Suspended particles	55
3.5.5.2	Organic matter	56
3.5.5.3	High alkalinity	56
3.5.5.4	Nitrates	57
3.6	Saccharides and high-intensity sweeteners	57
3.6.1	Bulk sweeteners	57
3.6.2	Intense sweeteners	58
3.7	Carbon dioxide	58
3.8	Acidulants	59
3.8.1	Citric acid	60
3.8.2	Tartaric acid	60
3.8.3	Phosphoric acid	61
3.8.4	Lactic acid	61
3.8.5	Acetic acid	62
3.8.6	Malic acid	62
3.8.7	Fumaric acid	62
3.8.8	Ascorbic acid	63
3.9	Flavourings	63
3.9.1	Flavourings and legislation	65
3.9.2	Flavourings in application	66
3.9.3	Water-miscible flavourings	66
3.9.3.1	Flavouring mixtures	66
3.9.3.2	Flavouring essence	67
3.9.3.3	Flavouring extract	68
3.9.4	Water-dispersible flavourings	68
3.9.4.1	Brominated vegetable oil (BVO) emulsions	72

3.10	Colours	73
3.11	Preservatives	75
3.11.1	Micro-organisms and beverages	76
3.11.2	Sulphur dioxide	77
3.11.3	Benzoic acid and benzoates	78
3.11.4	Sorbic acid and sorbates	79
3.12	More functional ingredients	79
3.12.1	Stabilisers	79
3.12.2	Saponins	80
3.12.3	Antioxidants	80
3.12.4	Calcium disodium EDTA	81
3.13	Retrospective investigation of a soft drink	81
3.14	Food safety	84
3.15	Future trends	85
	Bibliography	86

4 Syrup preparation and syrup room operations 87

John Horman

4.1	Introduction	87
4.2	Syrup composition	88
4.3	Syrup rooms and proportioning systems	90
4.4	The modern syrup room	91
4.5	Instrumentation	96
4.6	Sugar	97
4.6.1	Sugar dissolving	97
4.6.2	Liquid sugar storage	99
4.7	Pre-mixes	100
4.8	Pasteurisation	101
4.8.1	Tunnel pasteurisation	101
4.8.2	Flash pasteurisation	104
4.9	Clean-in-place (CIP) systems	105
4.10	Process plant hygiene requirements	108
4.11	Syrup room building design	109
4.12	Future developments	110

5 Carbon dioxide, carbonation and the principles of filling technology 112

David P. Steen

5.1	Introduction	112
5.2	Carbon dioxide	113
5.3	Production of carbon dioxide	115
5.3.1	Fermentation	116

5.3.2	Flue gas recovery	116
5.3.3	Membrane separation systems	118
5.4	Specification of carbon dioxide	119
5.5	Delivery to the customer	120
5.6	Carbon dioxide storage	123
5.7	Carbonation	124
5.7.1	Basic considerations	124
5.7.2	Carbonation measurement	126
5.7.3	De-aeration	128
5.8	Carbonation systems	129
5.8.1	Carbonators	129
5.8.2	Gas bubble sparging	131
5.8.3	Bubbles in a glass of carbonated liquid	133
5.8.4	Process control	133
5.9	Filling principles	133
5.9.1	Gravity filler	135
5.9.2	Counter-pressure filler	137
5.10	Nitrogen	142
	Further reading	142

6 Modern filling systems for carbonated soft drinks 144

Andrew Wilson and John Medling

6.1	Introduction	144
6.2	Modern carbonation systems	147
6.3	Counter-pressure fillers	149
6.3.1	Operation	149
6.3.2	Level filling	154
6.4	Glass bottle filling	156
6.4.1	Operation	156
6.4.2	Bottle burst protection	159
6.5	PET (polyethylene terephthalate) bottle filling	161
6.5.1	Operation	161
6.5.2	Size changing	162
6.6	Can filling	166
6.6.1	The volumetric can (VOC) filler	166
6.6.2	System expansion – the VOC-C can filler	168
6.6.3	Product level- and pressure-control system in a free-standing product tank	170
6.6.3.1	Level control	170
6.6.3.2	Filling pressure control	170
6.6.4	Procedural steps of the filling cycle	170
6.6.4.1	Flushing phase I	171
6.6.4.2	Flushing phase II	171

6.6.4.3	Pressurising	171
6.6.4.4	Filling	171
6.6.4.5	Snifting	172
6.6.4.6	Recharging the metering chamber	172
6.6.4.7	CIP cleaning	173
6.7	Other filler options	174
6.8	Clean-in-place	175
6.9	Filler hygienic design	177
6.10	Conclusions	178

7 Bottle design and manufacture and related packaging 181

David Syrett

7.1	Introduction – the objectives of packaging	181
7.2	Glass	182
7.2.1	History of glass making	182
7.2.2	Bottle supplier identification marks	184
7.2.3	Bottle design and headspace considerations	184
7.2.4	Glass bottle manufacturing processes	185
7.2.4.1	The melting process	185
7.2.4.2	Glass bottle forming	187
7.2.5	Glass quality faults	189
7.2.5.1	Critical defects	189
7.2.5.2	Major defects	189
7.2.5.3	Minor defects	190
7.2.6	Single and multiple trip bottles	192
7.2.7	Lightweighting	193
7.2.8	Recycling	194
7.3	Plastic bottles	194
7.3.1	Introduction – advantages and limitations	194
7.3.1.1	Advantages	194
7.3.1.2	Limitations	195
7.3.2	Polyethylene terephthalate	195
7.3.3	History	195
7.3.4	PET bottles today	196
7.3.4.1	Preforms	196
7.3.4.2	Bottle blowing	196
7.3.5	Multilayer and barrier coating	198
7.3.6	Carbonation loss	198
7.3.7	Permeability of polymers	198
7.3.8	Recycling methods	200
7.4	Closures for carbonated soft drinks	200
7.4.1	Introduction	200
7.4.2	Headspace pressure	201

7.4.3	Roll on metal closures	202
7.4.4	Plastic closures	202
7.4.5	Development of the PET bottle finish	203
7.4.6	Crown corks (pry off/twist off)	205
7.4.7	Other closures (vacuum lug/rip off)	205
7.5	Cans and ends	206
7.5.1	Introduction	206
7.5.2	The can making process	206
7.5.2.1	Front-end processes	206
7.5.2.2	Decoration processes	207
7.5.2.3	Back-end processes	207
7.5.3	Graphics decoration	208
7.5.4	Ends	208
7.5.5	The double seam	208
7.5.6	Can end stress corrosion	208
7.6	Labels and labelling	209
7.6.1	Introduction	209
7.6.2	Paper manufacturing	210
7.6.3	Labelling	210
7.6.4	Patch labellers	212
7.6.5	Wraparound labellers	213
7.6.6	Sleeving	215
7.6.7	Labelling in production	215
7.6.8	Date coding (ink jet/laser)	216
7.7	Future trends – putting it all together	216
	Further reading	217

8 Secondary packaging considerations 218

David Rose

8.1	Introduction	218
8.2	Requirements of the supply chain	219
8.2.1	Overview	219
8.2.2	Major multiple	221
8.2.2.1	Pallet receipt	221
8.2.2.2	Pallet breakdown and load picking	221
8.2.2.3	Bar codes	221
8.2.2.4	Store delivery	222
8.2.3	Cash & Carry	224
8.2.3.1	Pallet breakdown and load picking	224
8.2.3.2	Pack graphics	224
8.2.3.3	Bar codes	224
8.2.4	Bulk break	224
8.2.5	Poor packs – no sale?	225

8.2.6	Material – performance versus specification	225
8.3	Pack styles	227
8.3.1	Shrink wrap	227
8.3.2	Trade ready display unit (TRDU)	228
8.3.3	Shelf ready packing	229
8.3.4	Returnable crate	231
8.4	Shrink film production	231
8.4.1	Film considerations	231
8.4.2	Shrink wrap packers	233
8.4.3	Printed shrink wrap	234
8.5	Corrugated fibreboard	239
8.5.1	Specifications	239
8.5.2	Corrugated board manufacture	240
8.6	Stretch wrap	240
8.6.1	Material	240
8.6.2	Stretch wrapping systems	244
8.6.3	Pallet wrapping	247
8.7	Pallets and pallet stacking	248
8.7.1	Pallets	248
8.7.2	Pallet stacking	248
8.8	Case and pallet labelling	249
8.9	Getting it wrong	253
8.10	Developments and testing	257
8.11	Putting it all together	258

9 Production systems 259

David Steen

9.1	Introduction	259
9.2	Philosophy of line layouts	259
9.3	Glass bottle filling lines	260
9.4	Can filling lines	267
9.5	PET bottle filling lines	268
9.6	Conveying systems	270
9.7	Inspection systems	272
9.8	Breakdown characteristics	274
9.9	Line control and management systems	277
9.10	Complete factory layouts	280
9.11	Buildings	283
9.12	Performance measurement and benchmarking	285
9.13	Future trends	288
	Further reading	289

10 Production planning and distribution 290

Philip A. Wood

10.1	Introduction	290
10.2	Supply chain principles	291
10.2.1	Make versus buy	292
10.2.2	Make to order or make to stock	293
10.2.3	Purchasing	294
10.2.4	The soft drinks supply chain	296
10.2.5	The bullwhip effect	298
10.3	Forecasting	299
10.3.1	Forecast breakdown or types	300
10.3.2	Forecast preparation and accuracy	301
10.4	Planning	301
10.4.1	The planning and order cycle	301
10.4.2	Central or local planning	304
10.5	Warehousing	307
10.5.1	Environment	314
10.5.2	Stock control	314
10.6	Transport and legislation	315
10.7	Measuring supply chain performance	316
10.7.1	Service	316
10.7.2	Costs	317
10.7.3	People	317
10.7.4	Resources	317
10.8	Conclusions and future developments	317
	Bibliography	318

11 Quality, environment and food safety systems 319

Ray Helliwell

11.1	Why have systems?	319
11.1.1	Definitions	319
11.1.2	Space shuttle fails	319
11.1.3	Risk management	320
11.1.4	A solution?	320
11.2	Management control	321
11.2.1	Constraints – legislation	321
11.2.2	Stakeholders determine your future	321
11.2.3	Size matters	322
11.2.4	Experts, but not owners	322
11.3	Management systems	323
11.3.1	Four stages	323
11.3.2	Decisions	323

11.3.3	Make the product	323
11.3.4	Test and record results	324
11.3.5	Review	324
11.4	International management systems – why?	325
11.4.1	Notice board systems	325
11.4.2	Detect and protect	325
11.4.3	Automation = automatically wrong	326
11.4.4	Quantity and quality	326
11.4.5	Systems converge	327
11.5	The standards – a brief background	327
11.5.1	ISO 9001 is the master	327
11.5.2	Customer pressure	327
11.5.3	Customer audits	328
11.5.4	A standard is born	328
11.5.5	Is ISO 22000 necessary?	329
11.5.6	Other standards	330
11.5.7	Systems director	330
11.5.8	Standards improvement mechanisms	331
11.6	ISO Standards – common elements	331
11.7	The quality management system (QMS) – BS EN ISO 9001	332
11.7.1	Resource management [6]	332
11.7.2	Contract review [7.2]	332
11.7.3	Purchasing [7.4]	332
11.7.4	Product realization [7]	332
11.7.5	Product development [7.3]	333
11.7.6	Control or test [7.5.2]	333
11.7.7	Quality assurance – process control	333
11.7.8	Quality control – sample testing	334
11.7.9	Warehousing [7.5.5] and distribution [7.5.1]	335
11.7.10	Control of non-conforming product [8.3]	335
11.8	The food safety management system (FSMS) – BS EN ISO 22000	335
11.8.1	The standard of the future?	335
11.8.2	Policy [5.2]	335
11.8.3	Pre-requisite programmes [7.2]	336
11.8.4	Product characteristics [7.3.3]	336
11.8.5	HACCP [7.6]	336
11.8.6	Decision tree	337
11.8.7	No foolproof systems	339
11.9	The environmental management system (EMS) – BS EN ISO 14001	339
11.9.1	Drains and skips – duty of care	339
11.9.2	Producer responsibility	339
11.9.3	Climate change levy	340
11.9.4	Pollution prevention and control permits	340
11.9.5	EMS policy [4.2]	340
11.9.6	Aspects and impacts [4.3.1]	340

11.9.7	Significance [4.3.1(b)]	341
11.9.8	Legislation updates [4.3.2]	341
11.9.9	Emergency preparedness and response [4.4.7]	341
11.10	Systems – a summary	342
Index		343

Colour plate section appears after page 174

List of Contributors

Dr Philip Ashurst	Dr P. R. Ashurst & Associates, Unit 5, Gooses Foot Estate, Kingstone, Hereford, HR2 9HY, UK
Dr Bob Hargitt	British Soft Drinks Association, 20–22 Stukeley Street, London WC2B 5LR, UK
Mr Ray D. Helliwell	A.G. Barr plc, Mansell House, Aspinall Close, Middlebrook, Horwich, Bolton, BL6 6QQ, Lancashire, UK
Mr John Horman	Diessel Metering Systems Ltd., Cross Edge, Green Haworth, Accrington, Lancashire, BB5 3SD, UK
Mr John Medling	Krones UK Ltd., Westregen House, Great Bank Road, Wingates Industrial Park, Westhoughton, Bolton, Lancashire, BL5 3XB, UK
Mr David Rose	Britvic Soft Drinks Ltd., Widford, Chelmsford, CM1 3BG, UK
Dr David P. Steen	Casa Davann, Las Rellanas 26-F, Arboleas 04660, Almeria, Spain
Mr David Syrett	Britvic Soft Drinks Ltd., Widford, Chelmsford, UK
Mr Walter Tatlock	CBJ Service Ltd., 17 Kirkfell Drive, Astley, Tyldesley, Manchester, M29 7GB, UK
Mr Barry Taylor	Dansico Ingredients Ltd., Denington Estate, Wellingborough, NN8 2QJ, UK
Mr Andrew Wilson	Krones UK Ltd., Westregen House, Great Bank Road, Wingates Industrial Park, Westhoughton, Bolton, Lancashire, BL5 3XB, UK
Mr Philip A. Wood	A.G. Barr plc, Mansell House, Aspinall Close, Middlebrook, Horwich, Bolton, BL6 6QQ, Lancashire, UK

Preface

The market for carbonated beverages has grown dramatically in most countries, for example, by 128% in the UK since 1984. This growth has required changes in the way factories are run. Soft drinks are now classified as food products and are produced under stringent hygiene conditions. Twenty years ago this was not always the case – carbonated drinks were often produced in old buildings where cross-contamination could easily occur. Filling technology has progressed rapidly to meet the needs of manufacturers and consumers alike. Whilst the basic counter pressure filler is still the main work horse, new generations of electronically and pneumatically controlled filling machines have been developed that allow production under much more hygienic conditions and to much higher standards of filling accuracy and repeatability.

Primary packaging for carbonated soft drinks in most countries appears to have settled on PET bottles, with 500 ml and 2 l sizes predominating. The returnable glass bottle business is in decline and single-trip glass bottles are now mainly used for specialist products. The growth of can use appears to have stabilised. The 'lightweighting' of PET bottles over the last decade appears to be reaching its limits. Work on this will inevitably continue though at a slower rate as a consequence of the high cost of oil. Developments in barrier technology and new resins mean that shelf life in small PET bottles is not the problem it once was for carbonated products, although the much higher bottle cost restricts the use of such materials to premium product. Closure technology has also improved over the last 10 years with the advent of new neck finishes for PET bottles and improvements in closure design.

After an introduction to carbonated soft drinks, this volume considers the specifications and treatments for water, the main ingredient of any soft drink. A chapter on the other ingredients and formulation of carbonated drinks then follows (Chapter 3). Chapter 4 considers how ingredients are blended together to an agreed specification, and includes sections on sugar dissolving and batch and continuous syrup production. The predominant philosophy now is to manufacture the finished product as ready to drink (RTD), and prove that it is within specification prior to carbonation. In this way losses are minimised, uniformity of product is guaranteed and there is tight cost control of production.

A chapter on carbon dioxide production and the physics of carbonation then follows (Chapter 5). Modern carbonation techniques and feedback control are considered, with a discussion of different carbonation methods. The physics of filling carbonated beverages is included, followed by a consideration of the latest generation of filling machines (Chapter 6).

Chapter 7 deals with primary packaging – the concepts of containing, protecting, identifying and marketing are considered in the light of environmental, legislative and cost factors for the three main containers: glass bottles, PET bottles and cans.

Chapter 8 is on secondary packaging, explaining its importance during the development of a new or existing product. The function of secondary packaging is to ensure that the primary container is delivered to the consumer in prime condition, at the same time as satisfying the ever increasing demands of the trader.

A further chapter deals with production systems, applying the topics of previous chapters to the production line and factory, discussing glass, PET and can lines and describing principles and inspection systems (Chapter 9). This chapter also considers plant breakdown characteristics, line control and management systems. An introduction to the requirements for factory layouts and design is then followed by considerations of performance measurement and benchmarking.

Chapter 10 deals with the increasingly important subject of production planning and distribution. As a consequence of the high weight and comparatively low value of carbonated soft drinks, this topic is receiving much more attention than previously. Supply chain management is discussed in relation to soft drinks, highlighting its importance. It is no longer enough to just produce soft drinks, they must be produced uniformly every day and be distributed to the customer at the lowest possible cost if the producer is to stay in business. Chapter 11 is on quality, environment and food safety; completing the picture by providing the framework within which manufacturing and distribution must now exist.

The aim of this volume is to provide an overview of carbonated soft drinks production in the early part of the twenty-first century, presenting the latest information on carbonation and filling methods. Detailed references provide opportunity for further reading in more specialised areas. Certain topics, such as ingredients and packaging, are not included in great depth here because they are covered in detail elsewhere in the series. The book is aimed at graduates in food science, chemistry, microbiology and engineering who are considering a career in the soft drinks industry, as well as technical staff already employed within the industry and associated suppliers.

The editors are greatly indebted to the contributing authors: without them this book would not exist. All are experienced in their particular fields and, for most of them, the work involved in writing their chapters was a significant extra burden on top of their already heavy workload.

David Steen

Philip Ashurst

1 Introduction

Bob Hargitt

The aim of this introduction is to provide a brief perspective on the historical development of carbonated soft drinks from their first appearance in the late eighteenth century to the present day, and also to consider their future path.

1.1 Early history

The consumption of soft drinks in their various forms has taken place for many centuries in order to meet the body's fundamental requirement for hydration. The most obvious source of hydration is water, but in earlier times the consumption of water was very hazardous as it was frequently contaminated by micro-organisms. Outbreaks of cholera, dysentery and other waterborne illnesses were common in many European cities prior to the twentieth century. The consumption of 'small beers' – drinks which had been boiled, flavoured with, for example common herbs, and slightly fermented – was widespread. Barley waters, flavoured drinks containing pearled barley, were recorded as early as 1320 and the earliest English reference to lemonade was published in 1663. The drink contained lemon juice and was sweetened with sugar or honey and is thought to have originated in Italy. Orangeade was also recorded in the 1660s. All these early drinks were, of course, not carbonated.

Production of effervescent alcoholic beverages, that is, beers and wines where the carbon dioxide was derived directly from fermentation, is recorded as beginning at the latest in 1693, when Dom Perignon is credited with the invention of champagne. However, references to sparkling wines are found in English literature well before this date. Several spas were also known where the water was naturally effervescent and during the seventeenth century scientific interest and study grew in the gas which caused this effect, particularly at Spa in Holland and Pymont and Seltzer in Germany. There was considerable scientific investigation across Europe of the gas we now know as carbon dioxide (CO_2) by the middle of the century. In 1741, Brownrigg termed the gas 'mephitic air' and produced effervescent waters from bicarbonate salts. By 1767, Richard Bewley was selling effervescent 'mephitic julep' for its medicinal properties. In 1764, Macbride in Ireland demonstrated the medicinal uses of effervescent waters and their antiseptic properties.

The discovery of the means of artificially carbonating water by dissolution of CO_2 under pressure is attributed to Dr Joseph Priestley in the late 1760s, though there were many other workers active in this field at the same time who probably deserve equal credit. He published his work, *Directions for Impregnating Water with*

Fixed Air in 1772 and Dr John Mervin Nooth developed an apparatus for preparation of effervescent waters, which he reported in the *Philosophical Transactions of the Royal Society* in 1775. Torbern Bergman, Professor of Chemistry at Uppsala University in Sweden, published his work on preparation of artificial mineral waters in 1773. His treatise on Bitter, Seltzer, Spa and Pyrmont Waters has been termed the world's first textbook on the manufacture of mineral waters. In 1780, Duchanoy in France published a treatise on the art of imitating naturally occurring mineral waters. The initial commercial development, deriving from all this scientific work, was that of selling imitation mineral waters, that is, waters to which were added minerals in the proportions found in naturally occurring mineral waters and then artificially carbonated. Hence the term in English of 'aerated mineral waters', which became synonymous with all carbonated drinks. The commercial development of carbonated waters took off very rapidly following the initial scientific and technical discovery.

Thomas Henry, a Manchester apothecary, is generally credited to have been the first commercial manufacturer of artificially carbonated water in the late 1770s. He improved and developed Nooth's design to make an apparatus capable of carbonating batches of up to 12 gal (54 l). The product was sold in tightly corked glass bottles. Henry recommended consumption of lemon juice and soda water for the stomach but did not state whether the two were combined. By the late 1770s he was also selling artificially manufactured Pyrmont and Seltzer waters, that is, imitations of the naturally occurring spa waters. Thomas's son, Dr William Henry, was the inventor of Henry's Law of Gases (1805). The storage of fresh water on board ships during long sea voyages, which could last many months, was a serious problem and the antiseptic effect of carbonation and hence its long 'shelf-life', made an immediate impact upon the British Admiralty. It was also claimed (wrongly) that soda water cured scurvy and one of the first uses of carbonated water was on board a ship. Bottles of soda water have been salvaged from the wreck of the 'Royal George' which sank in 1782. Early effervescent drinks, similar to Bewley's Mephitic Julep, were manufactured by mixing sodium bicarbonate solution with lemon juice or lime juice, which would of course cure scurvy. This was probably the cause of the misconception that CO₂ was a cure for scurvy.

The manufacture of carbonated drinks also rapidly became popular across Europe. Paul, Schwappe & Gosse established a successful business in Geneva, before Jacob Schwappe moved to London in 1792 to set up a factory-scale operation there. The production of mineral waters was well established by 1800, and J. Schwappe & Co opened another factory in Bristol in 1803. His former partner, Nicholas Paul also moved to London in 1802 and set up in competition with Schwappe. Paul is credited with the first commercial use of a high pressure gas pump to aid dissolution and achieve high levels of carbonation, his mineral waters were famous for containing several volumes of CO₂.

Carbonated waters were imported into the USA from the UK prior to 1800. The first commercial production is attributed to Benjamin Silliman, who was professor of chemistry at Yale College. He had seen carbonated waters on his travels to England and had met Joseph Priestley, who had emigrated to the USA. In partnership

with a Mr Twining, he began selling carbonated waters in 1807 in Newhaven. Joseph Hawkins established an enterprise in Philadelphia the same year and operations rapidly sprang up in other cities in the north-east, for example, New York, Baltimore and Boston.

An excellent account of the development of the soft drinks industry in the USA was written by John J. Riley.

1.2 The growth of carbonates – production

The carbonated soft drinks industry continued to grow steadily as the nineteenth century progressed. By 1840, there were at least 50 manufacturers in London. At the Great Exhibition, held in London in 1851, J. Schweppe & Co paid £5000 for the concession to sell ‘soda and other mineral waters’. They sold in excess of 1 million bottles during the course of the exhibition. Throughout the nineteenth century the popularity of carbonated soft drinks increased steadily and the number of flavours expanded likewise, driven by the popularity of the temperance movement. This growth of carbonates coincided with the industrial revolution through the nineteenth century. Production of soft drinks became more industrialised and a process of continuous improvement soon developed. The *Mineral Water Maker’s Manual for 1885* lists over 80 patents, which were registered in the previous year, related to the manufacture and packaging of bottled soft drinks. It also lists approximately 300 trademarks which had been approved between the passing of the Trade Marks Act in 1875 and 1881, including that for the Buxton Mineral Water Co., Fairfield Works, Buxton, in 1876.

Continuous improvement in production and packaging of carbonated soft drinks meant that by the middle of the nineteenth century a manual bottling line was capable of filling 100 dozen bottles per day, but the introduction of steam power increased that to 300 dozen per day. By 1900, it was estimated that 70,000 people were directly employed in the UK soft drinks industry and 22,000 horses were used for product delivery. Total UK production was estimated by Bratby & Hinchcliffe to be almost 300 million dozen half pints (ca. 900 million litres). For comparison, in 1990, government statistics show that almost 18,000 people were employed in the soft drinks industry (manufacturing, distribution, sales and marketing) producing 6717 million litres of drinks. In 1899, there were 2763 soft drink bottling plants in operation in the USA.

The industries of the UK, Europe and USA progressed along slightly different paths owing to the differing circumstances found in those regions, although three types of beverage were found in each region. The industry in the UK, which was becoming more industrialised with large factories supplying products to the masses, progressed along the path of industrial production of soft drinks in returnable bottles sold through shops. In continental Europe the soda siphon type device (i.e. the gasogene or seltzogene) became popular for home use. These were used for the dispensing off of flavoured drinks, not just soda water. The common soda siphon which we would recognise today was patented by Charles Plinth in 1813. However,

he used a stop-cock to dispense the contents and the use of a lever-operated device was patented by Deleuze and Dutillet in Paris in 1829. The use of a small metal bulb filled with CO₂ to re-charge a siphon of water was patented by Arthur Marescot in 1874.

In the USA, soda fountain equipment, where drinks were produced in shops for consumption on site, also became very popular. Some carbonates were consumed purely as a source of refreshment but many retained their medicinal pedigree to a greater or lesser extent. The most notable was probably quinine tonic water, which was consumed in tropical regions as a cure for malaria. Dandelion and burdock was obviously of herbal origin, and another popular drink in late nineteenth-century Scotland and in London during the 1890s was Kola Tonic. Kola (or cola) was a nut from West Africa, which was used by Nigerians as a symbol of hospitality. In 1886, Dr John S. Pemberton combined cola with coca (an extract from the S. American coca leaf) to produce his coca-cola 'brain tonic' sold in the soda fountain of his store in Georgia. In 1892, Asa G. Candler took over the business and incorporated the Coca-Cola Company in Atlanta with an aggressive marketing campaign for his 'nutrient beverage and tonic'. The company granted the rights to bottle the product under licence. The first such plant opened in Chattanooga in 1899, followed rapidly by many more. Around the same time Dr Pepper was launched by R.S. Lazenby in Waco, Texas (ca. 1888) and Pepsi-Cola was launched at New Bern, North Carolina by Caleb D. Bradham (ca. 1896, although the name Pepsi-Cola was not coined until 1901).

By the close of the nineteenth century most of the common carbonated soft drinks of today were already on sale, for example, soda water, ginger beer, ginger ale, lemonade, orangeade and other citrus drinks, cherryade, quinine tonic water, bitter lemon, colas, sarsaparilla, root beer, cream soda etc. These would all have been well known to consumers in the late Victorian era.

There is a difference between the American and British definitions of soda water. In the USA, soda water is defined simply as carbonated water, but the UK legislation still requires that soda water must contain a minimum of 550 mg/l sodium bicarbonate. Interestingly the only other legal compositional standard for a soft drink in the UK is for tonic water, which currently must contain a minimum of 57 mg/l quinine (as sulphate).

1.3 Technological development

1.3.1 Carbon dioxide

It had been recognised by many scientists in the early 1700s that the gas produced by brewery fermentation, combustion of wood and addition of acids to chalk/marble was one and the same. It was given several names including artificial air (Boyle 1685), mephitic air (Brownrigg 1741), fixed air (Black 1754), gas acide carbonique (Lavoisier 1782) and finally gaz oxide de carbon (Fourcroy 1805).

The most practical/economic means of commercial production was by the action of sulphuric acid on marble chippings (known as whitening) or, at a later date, on sodium bicarbonate. Crushed marble (or chalk or limestone) was cheap and readily available in large quantities. However, the purity of the marble was critical to the quality of the CO₂. Impurities (particularly organic ones) would cause noticeable 'off flavours' in the finished drink. This forced manufacturers to introduce filters and scrubbers to remove taints. Bubbling the CO₂ through olive oil was a commonly used method of removing organic taints. The purification of CO₂ introduced complexity and hence cost to the production process. Although more expensive than marble, sodium bicarbonate could be obtained in commercial quantities at consistently high purity and was preferred by some manufacturers. The product of the action of sulphuric acid on marble is calcium sulphate, which is insoluble in water. Large quantities of the resulting sludge were difficult to dispose of, particularly when the UK municipal authorities introduced controls in the 1890s. Problems of effluent emissions are not new. The liquification of CO₂ by means of high pressure was reported by Michael Faraday in 1823 and the first practical manufacturing equipment was patented by Dr Henryk Beins in Holland in 1877. The commercial manufacture and use of liquid CO₂ for the carbonation of drinks began in Germany and in the USA in the 1880s. The production of solid CO₂ was discovered by Thilorier in 1835, and a patent for the production and use of solid CO₂ was granted to Dr Samuel Elworthy in 1897. The handling and transportation of solid blocks of CO₂ was much easier than for heavy metal cylinders containing liquid CO₂. Though use of liquid or solid CO₂ increased in the late nineteenth and early twentieth century, it was not until the 1950s that transportation of liquid CO₂ by low pressure bulk road tankers became commonplace.

Production of carbonated drinks was traditionally carried out by means of adding concentrated syrup to the bottle and then topping up with carbonated water. A considerable improvement in speed was achieved in 1937, when the Mojonner Brothers Corporation of Chicago introduced a continuous blending/cooling/carbonating system. Such continuous systems have gradually replaced the syrup dosing systems, though some of the latter remained in operation into the 1980s.

1.3.2 Sweeteners

One of the major drivers of growth in carbonates has been the development of sweeteners and consequent improvement in the quality of low calorie soft drinks, particularly in the USA and UK. Saccharin was invented in about 1874 and very rapidly became popular as a sweetener for soft drinks, usually blended into sugar to reduce cost. Figure 1.1 shows an advertisement for saccharin, which was published in 1890, by which time it was already widely used. A modern supplier may have difficulty substantiating all of the claims made for it. However, it proved to be a popular sweetener in the UK, particularly when sugar was in very short supply during the First World War. A blend of sugar and saccharin (50 : 50 by sweetness) became the standard sweetener system for common soft drinks, for example, lemonade.

- Impulse sector, 298
- Infiltration, 17
- Ingredients, 48, 88, 90, 100
 - room, 109
- Inkjet coding, 216
- Innovative products, 293, 294
- Inspection systems, 272–4
- Instrumentation, 96–7, 123, 141
- Integrated pollution control legislation, 22
- International management systems, 325–7
- International Standards Organisation (ISO), 327
- Kalsep Fibrotex unit, 40
- Label paper manufacture, 210
- Labels, 209–16
- Labelling, 210–12
 - cases and pallets, 249–53
 - patch, 212–13
 - PET bottles, 269
 - plastic, 213–15
 - in production, 215–16
 - wraparound, 213–15
- Laser coding, 216
- Lead time, 290, 296, 299, 301, 307, 311, 313, 316
- Legislation, 315, 321, 339, 341
- Lemonade, 1
- Level filling, 154–6
- Lighting, 274, 284
- Line
 - control, 277–80
 - layouts, 259–60, 283
- Logistics, 290, 291, 296, 300
- Low calorie carbonates, growth of, 7
- Major multiple retailers, 298, 317
- Management
 - control, 321–2
 - decisions, 323
 - Information Systems (MIS), 278
 - systems, 277–80, 323–5
- Manufacturing resources planning (MRPII), 301
- Material performance, 225
- Materials requirements planning (MRP), 301
- Mechanical handling equipment (MHE), 307, 309
- Microbial growth in poor pipe welds, 87
- Micro-organisms and beverages, 76, 338
- Mineral waters naturally occurring, 2
- Mojonnier system, 91
- National distribution centre (NDC), 305
- Neck lifters, 162
- Nephelometer, 28
- Nitrates, 25, 36, 57
- Nitrogen, 36, 142
- Non-conforming product control, 335
- Notice board systems, 325
- Nucleation, 43, 124
- Nutritional fortification of drinks, 13
- Occupational exposure standard, 115
- Occupational Health and Safety Management System (OHSAS 18001), 330
- Order cycle, 301
- Overall effective efficiency (OEE), 275, 285, 288
- Oxygen pick up, 145
- Ozone technology, 42–3
- Pack
 - conveyors, 272
 - graphics, 224, 228, 238
- Packaging
 - collation, 218
 - early types, 8, 51
 - objectives, 181–2
 - secondary, 218–58
- Pallet
 - breakdown and load picking, 221, 224
 - conveyors, 272
 - receipt, 221, 224
 - stacking, 240, 248–9
 - wrapping, 247–8
- Palletisation problems, 219, 253–4
- Palletised goods, standard test criteria, 258
- Palletiser, robot, 264
- Pallets, 248
- Partial pressure, 126–8, 134
- Pasteurisation, 99, 101–5
 - convection flow, 102
 - flash, 104–5
 - tunnel, 101–3
- Performance, 288
 - measurement, 285–8
- PET, *see* Polyethylene terephthalate

- PET bottle
 - biaxial orientation, 197
 - blowing, 196–7
 - filling, 145, 161–6
 - lines, 268
 - permeability, 198
 - silos, 269
 - stretch blow moulding, 269
- PET performance, 196
 - multilayer and barrier, 198
- PET recycling, 200
- Photosynthesis, 114
- Pig systems, 94–6
- Planning, 301–6
- Plant efficiency, 285
- Plastic
 - bottles, 194–200
 - limitations, 195
 - crates, 231, 260
- Pollution prevention and control, 340
- Polyethylene terephthalate (PET), 195–6
- Poor packaging, 219, 220, 225, 238, 240, 254
- Pre-mixes, 100
- Preservatives, 74
- Pressure sustaining valve, 45
- Primary packaging, 181–217
- Procedures, 314, 319, 323, 325, 337
- Process
 - control, 133–4, 332–3
 - plant hygiene requirements, 108
- Producer responsibility, 339–40
- Product
 - ageing, 314
 - characteristics, 336
 - development, 293, 294, 296, 300, 333
 - quality, 285
 - realisation, 332–3
- Production
 - availability, 288
 - cost models, 285
 - ‘just in time’, 305
 - plant commissioning, 286
- Production line
 - breakdown characteristics, 274
 - simulation, 275
- Proportioning systems, 90–1
- Purchasing, 294–6, 301, 332
- Quality, 288, 290, 294, 326
 - assurance, 333
 - control, 268, 282, 334
- Quality management system, QMS (ISO 9001) 327, 332–5
- Quinine tonic water, 4
- Radio frequency identification (RFID), 252, 317
- Ready to drink (RTD), 112
- Refractometer, 96–7
- Refractometric solids content, 96
- Refrigeration unit, 121, 123, 129
- Refrigerated carbonation system, 129
- Regional distribution centres, 220
- Regulations, UK
 - Environmental Protection (Duty of Care), 339
 - Pollution, Prevention and Control, 321, 340
 - pressure equipment, 40
 - Producer Responsibility (Packaging Waste), 339
 - Water, 22
 - Water Supply, 19
 - Weights and Measures (Packaged Goods), 272
- Resource management, 332
- Resources, 295, 296, 315, 317, 323, 330
- Returnable crates, 231
- Reverse osmosis, 31
- Risk management, 320
- Robot, 263, 270
- Saccharin, 5
- Sample testing, 334
- Sanitisation, 338
- Saponins, 80
- School drink restrictions, 13
- Sealing devices, 9, 51
- Secondary packaging, 218–58
 - test methods documentation, 257–8
 - transport simulation, 257–8
- Seltzer waters, 49
- Shelf ready packing, 220, 222, 228, 229–30
- Shrink
 - film production, 221, 230, 231–8
 - considerations, 231–3
 - wrap, 227–8
 - packers, 234
 - printed, 234–8
- Silt density index, 39
- Size changing, 161–6
- Slat lube, 271
- Sleeving, 215

- Sniffing, 137, 139, 141, 145, 154, 166, 167, 172
- Soda
 - fountain, 4, 49
 - siphon, 3
 - water, 4, 49
- Soft drinks
 - constituents, 52
 - future trends, 85
 - investigations, 81
 - supply chain, 296
- Solubility of CO₂ in water, 126–7, 149
- Sorbic acid and sorbates, 79
- Stabilisers, 79
- Stakeholders, 321–2
- Stock
 - control, 314
 - keeping units (SKU), 220, 251
- Store delivery major multiples, 222–4
- Stretch
 - wrap, 240–8
 - wrap material, 240–4
 - wrapping systems, 244–7
- Sugar
 - dissolving, batch, 97–8
 - continuous, 98–9
 - liquid storage, 99
 - tanker unloading, 98
- Sulphur dioxide, 77
- Supervisory control and data acquisition (SCADA), 278
- Supplier
 - development, 294, 295
 - selection, 294
 - shipping container code, 250
- Supply chain, 219–27, 270, 290, 292, 293, 296–8, 303, 339
 - costs, 290, 316
 - management, 291, 298
 - people, 316
 - performance measurement, 315–17
 - principles, 291–9
 - requirements, 219
 - resources, 317
 - service, 316
- Supply dual sourcing, 295
- Sweeteners and sweetener developments, 5, 57
- Syrup
 - composition, 88–90
 - room building design, 109–10
- Systems definitions, 319
- Technological developments, 10, 110–11, 112, 179
- Tolerable negative errors (TNE), 273
- Total planned maintenance, 287
- Trade
 - ready display units, 228–9
 - sector, 298
- Traded unit size, 220
- Transport, 293, 298, 300, 301, 303, 315
- Trend analysis, 287
- Turbidity, 19–23, 27–8, 32
 - meter, 19, 21–2
- Valve matrix, 93–5
- Vendor managed inventory, 305
- Warehouse
 - bulk break, 224
 - health and safety issues, 315
 - types, 307
- Warehousing, 282, 306–15, 335
- Wash facilities, 284
- Water
 - alkalinity reduction, 28–9
 - chlorination, 25, 32, 34–5
 - cycle, 16
 - daily checks, 21
 - electrodialysis, 37
 - framework directive, 23
 - hardness, 28, 55
 - impurities, 55
 - iron removal, 35–6
 - natural mineral, 18, 23
 - nitrate removal, 36–8
 - ozone technology, 42–3
 - quality, 18–21, 55
 - sand filtration, 25–7
 - softener, 33
 - sources, 18
 - spring, 18, 23
 - standards, 20
 - supply regulations, 19
 - table, 18
 - town, 21, 39
 - treatment, 16–47, 55
 - ultra violet disinfection, 25, 41
- Waterborne illnesses, 1
- Weighting agents, 72
- Weights and Measures Act, 272
- Windows, 284
- World class manufacturing, 285, 288
- X-ray systems, 274

